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Next item →

1. If A, B, C , and D are the state-space matrices of a continuous-time model, and A_d, B_d, C_d , and D_d are the corresponding discrete-time state-space matrices, which of the following statements are true in general? (Select all that apply.)

1 / 1 point

☐ $A_d = A$.

☐ $B_d = B$.

☒ $C_d = C$.

👉 **Correct**
Yes. This is correct.

☒ $D_d = D$.

👉 **Correct**
Yes. This is correct.

2. If A is the system matrix of a continuous-time model, how can we compute the corresponding A_d of a discrete-time model of the same system? Assume that the sampling period is Δt and select all that apply.

1 / 1 point

☐ $A_d = A$.

☐ $A_d = \exp(A * \Delta t)$.

☒ $A_d = \expm(A * \Delta t)$.

👉 **Correct**
Yes. This is correct.

☐ $A_d = I + A\Delta t$.

3. If A and B are the system matrix and input matrix (respectively) of a continuous-time model, how can we compute the corresponding B_d of a discrete-time model of the same system? Assume that the A matrix is invertible, that the sampling period is Δt , and select all that apply.

1 / 1 point

☐ $B_d = B$.

☒ $B_d = \int_0^{\Delta t} e^{A\sigma} B d\sigma$.

👉 **Correct**
Yes. This is correct.

☒ $B_d = A^{-1} (A_d - I) B$.

👉 **Correct**
Yes. This is correct. But, be careful... it requires that A be invertible, which is not always true.

☐ $B_d = B\Delta t$.